Python Programming– A Practical Approach

- **Pedagogy:**

  - Problem-Based Learning
  - Demonstration and Live Coding
  - Collaborative Learning.
  - Hands-On Experiential Learning

- **Learning outcome:**

  - Create and work with Python classes, objects, attributes, and methods.
  - Implement event handling and GUI components
  - Develop CRUD-based REST APIs using Python.
  - Perform basic hyperparameter tuning.
  - Develop an AI-powered resume screening solution.

Detailed Content:

Day 1 – OOPS using python

Topics

Forenoon Session

- Defining classes, instances, attributes vs methods
- Instance vs class attributes, __init__, self-keyword
- Encapsulation: private/protected attributes (name mangling, convention)
- Data hiding and validation inside classes.

Afternoon Session

- Inheritance and subclassing (super(), method overriding
- Polymorphism (duck typing, method overriding, abstract interfaces)
- Multiple inheritance, mix-ins, method resolution order (MRO), and how to use super() correctly in complex hierarchies.

Day 2 – GUI Programming with Python

Topics

Forenoon Session

- Introduction to GUI frameworks
- GUI frameworks: Tkinter, PyQt, Kivy
- Event-driven programming

Afternoon Session

- Creating multi-threaded GUI apps
- Integrating GUI with databases and APIs
- Packaging and distributing desktop applications

Hands-On Exercises:

Build a basic GUI calculator using Tkinter.

Create a file manager with PyQt to browse, rename, and delete files.

Develop a to-do list app with SQLite database integration.

Capstone Project:

Desktop Inventory Management System – A full-fledged inventory app with search, add, delete, and exportfunctionalities, using PyQt and SQLite.

Day3- Advanced Web Development with Python (Fast API & Flask)

Forenoon Session

- Introduction to Web Services
- Web frameworks: Flask vs. Fast API
- RESTful API development

Afternoon Session

- Authentication & security (JWT, OAuth)
- WebSockets for real-time applications
- Asynchronous APIs with Fast API

Hands-On Exercises:

Create a CRUD API with Fast API to manage user data. Implement JWT

authentication in a Flask web app.

Build a real-time chat application using WebSocket's in Fast API.

Capstone Project:

AI-Powered Resume Screening API – A Python-based REST API that receives resumes,

extracts key skills, and scores candidates using an AI model.

Day 4 – Machine Learning Pipeline Foundations

Topics

Forenoon Session

- Build, evaluate and serialize an ML model
- Data loading and exploration using Pandas and NumPy.
- Preprocessing and feature engineering (handling missing values, encoding categorical variables, scaling
  or transformation).

Afternoon Session

- Model selection: pick a simple supervised learner (e.g. logistic regression, decision tree, random forest,or a basic neural network if time allows).
- Train / test split, cross-validation, evaluation metrics, hyperparameter tuning.
- Serialize the trained model and preprocessing pipeline into a file

Day 5- Machine earning & AI Automation with Python

Forenoon Session

- Introduction to AI Automation
- Automating ML workflows with Python
- Training and deploying ML models

Afternoon Session

- AI-based automation (chatbots, recommendation systems)
- Using TensorFlow, PyTorch, and Scikit-learn
- Deploying AI models using Flask or Fast API

Hands-On Exercises:

Train a spam detection model and deploy it as an API.

Develop a real-time chatbot using NLP and OpenAI's GPT

API. Build an AI-powered resume screener for job

applications.

Capstone Project:

AI-Powered Customer Support Chatbot – A Python-based chatbot that uses NLP and deep learning to provide automated customer support.